ADVANCED BREAST CANCER DIAGNOSTIC RESEARCH PROTOTYPE

Formally Frozen V8 Architecture: Adaptive Wiener Denoising + Dual-Resolution U-KAN + Multi-Scale ViT (Arm F)

INPUT

1. CBIS-DDSM FFDM

Screening / Diagnostic
16-bit / 8-bit Gray Canvas
235 Quarantined Patients

DENOISE

2. Spatial Wiener + CLAHE

5x5 Adaptive Local Variance
Noise Mottle Attenuation
CLAHE Glandular Contrast
Min-Max Norm [0.0, 1.0]

U-KAN

3. Dual-Res U-KAN

512x512 + 768x768 Streams
Kolmogorov-Arnold Splines
Learnable 1D B-Splines
Frozen Arm G Checkpoint

TOP-K

4. Deterministic Proposals

Machine CC (tau=0.15, A>=15)
Prob Mass + Peak Scoring
Spatial NMS (IoU <= 0.65)
Top K=3 Candidates

STAGE 5: MULTI-SCALE RECEPTIVE FIELD GENERATION & CANDIDATE CROPS (K=3 PROPOSALS)

Local Scale (1.2x)

Lesion Core &
Microcalc Interior

Medium Scale (1.8x)

Perilesional Margins
& Micro-lobulation

Large Scale (2.5x)

Spiculation Radius
& Parenchymal Distort

Context Scale (4.0x)

Regional Background
Parenchymal Density

Calcification Patch

High-Pass Laplacian
Filter (224x224)

Global Mammogram

Full Breast Canvas
Context (224x224)

STAGE 6: V8 ARM F CANDIDATE-FUSION VISION TRANSFORMER (vit_base_patch16_224)

4x768-d

Candidate Attention

CandidateSoftmaxAggregator
alpha_k = Softmax(W^T e_k)
Pools K=3 -> 1x768 per scale

TUNED

Shared ViT Backbone

12 Transformer Blocks
Last 4 Blocks Fine-Tuned (8-11)
Blocks 0-7 Frozen

4608-d

Joint Latent Fusion

Local + Med + Lrg + Ctx (3072)
+ Calc (768) + Global (768)
Total Dimension: 4,608-d

Deep MLP Head: LayerNorm(4608) -> Linear(1024) -> GELU -> Dropout(0.25) -> Linear(512) -> Linear(128) -> Linear(2)

LOCKED V8

7. Research Output

P(Malignant) vs P(Benign)
Locked Acc: 64.57%
Locked F1: 64.18%
ROC-AUC: 0.6947
Sens: 66.05% | Spec: 63.55%

12-Layer Attention Rollout
Interpretability Saliency

MANDATORY DATA GOVERNANCE & PROTOCOL INTEGRITY:

- Primary Test Cohort: Exactly 525 scans / 235 quarantined patients with zero patient overlap between train, validation, and test splits.
- Checkpoint Provenance: models/final_v8/vit_v8_final.pth (MD5: 882313ce67adca01bffa699a180719c4). Zero post-hoc test retuning or retraining.
- Scientific Status: V8 is the permanently frozen final reported model. V9 is retained as an experimental extension. Academic research prototype only.